```
# Function to generate a title from a document
def generate_title(doc):
    bow = dictionary.doc2bow(doc)  # Convert document to bag-of-words
    topic_distribution = lda_model.get_document_topics(bow)
# Get topic distribution
    dominant_topic = max(topic_distribution, key=lambda x: x[1])[0]  # Select most probable topic
    # Extract top words from the dominant topic
    top_words = [word for word, _ in lda_model.show_topic(dominant_topic, topn=3)]
    # Generate a title by joining top words
    title = " ".join(top_words).title()
    return title
# Generate a title for a new document
new_doc = text[0]
generated_title = generate_title(new_doc)
print("Generated Title:", generated_title)
```

The output is:

```
Topic 0: [('language', 0.02 8249461), ('natural', 0.028061535), ('processing', 0.028007403), ('communication', 0.027926473), ('messaging', 0.027820466), ('human', 0.027805243), ('computers', 0.02779292), ('purpose', 0.027780347), ('instantly', 0.02777591), ('emails', 0.027773501)]
Topic 1: [('language', 0.07260038), ('natural', 0.056469582), ('human', 0.040358953), ('communication', 0.0403232), ('processing', 0.040299334), ('modify', 0.02420806), ('calls', 0.024204206), ('sentiment', 0.024201188), ('understand', 0.024201008), ('enables', 0.024199806)]
Generated Title: Language Natural Human
```

Topic modelling identifies recurring topics in a set of documents. This unsupervised model offers a simple and effective starting point for quick text classification. In order to determine the most likely title for a given text, it generates a set of tokens and their frequency from a corpus. Proper corpus training and word collecting are essential for success. The number of desired topics should also be minimal so that the target text is covered inclusively. In contrast to the other two methods, this extractive model is simple to comprehend and can be implemented with minimal effort using Python. I hope this post will inspire NLP fans to delve further in this domain. END

By: Dr Dipankar Ray

The author has more than 30 years of experience in software development. He has published two books on MATLAB and medical image processing (published by PHI Learning and SPD).